Manan Amit Shah

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EDUCATION

Indiana University Bloomington, IN, USA **Aug 2024 – May 2026**
Master of Science in Computer Science | **GPA: 3.76/4.00**
Coursework: *Applied Machine Learning, Applied Algorithms, Software Engineering I, Computer Vision, Computer Networks*

NMIMS University, Mumbai, India **Jul 2020 – May 2024**
Bachelor of Technology in Artificial Intelligence | **GPA: 3.41/4.00**
Coursework: *Data Structures and Algorithms, Object Oriented Programming, Operating Systems, Database Management Systems*

EXPERIENCE

Software Developer | *MyEdMaster, LLC* **Jun 2026 – Present**

- Architect and build the ed-tech startup's mobile app from scratch with **Claude Code** as an AI coding agent, producing **iOS** and **Android** builds of **9** screens from one **React** and Capacitor codebase as the sole engineer on the app
- Map **9** legacy server flows in a monolithic **Django** and **MySQL** backend with the web and interface teams and redesign them as stateless **JSON REST** endpoints, so web and native clients read the same records instead of drifting copies
- Document the new **10**-endpoint REST API in an **OpenAPI** specification and validate its endpoints with **Postman** collections, giving each endpoint a written request and response contract and a repeatable test
- Deliver token-based **authentication** for the mobile client with **Django REST Framework** and **SimpleJWT** through **5** endpoints for login, registration, logout, preference setup and account retrieval, replacing browser session auth

Machine Learning Engineer Intern | *STAT Institute* **May 2023 – Oct 2023**

- Engineered a scalable **microservices** backend with **REST & GraphQL** APIs for **ONNX** model deployment across distributed edge devices, using **Redis** caching to cut inference latency to **195ms** for the quantitative research team
- Devised an event-driven forecasting microservice around a **Temporal Fusion Transformer** model, feeding it multi-horizon time series inputs instead of the team's prior single-horizon baseline and improving **7**-day prediction accuracy by **35%**
- Diagnosed infrastructure bottlenecks on **AWS EC2** and configured custom **CloudWatch** alarm thresholds tuned to observed failure patterns instead of platform defaults, cutting false-positive alerts **19%** and easing on-call load

PROJECTS

Agentic Resume and Application Pipeline
Claude Code, MCP, Python, LaTeX, Bash

- Orchestrated a Claude Code agent workflow that takes each job posting through a viability gate, match scoring, drafting and scored critique rounds, using a steering document, **6** slash commands, per-job subagents and an **MCP** link to the Tsenta job platform
- Gated the agent with allow, ask and deny permissions, a **473**-line **rule checker** that rejects drafts with unverified numbers, forbidden terms, length errors or a second page, and an **approval step** so nothing is submitted without my explicit yes

Cutoff: Deterministic CPU Regression Engine
C++, Verilator, RISC-V, CMake, GoogleTest, AddressSanitizer

- Designed a **C++** engine that runs generated **RISC-V** programs on a reference model and a **Verilator** CPU simulation, checking **1,000,000** programs with zero mismatches, built against a staged spec with test gates and moved to **OpenAI Codex** midway
- Parallelized test runs across **8** worker processes using Unix fork and binary pipes, with a parent process that reorders results by program ID, reaching **22,220** programs per second, **5.82x** faster than one worker yet byte-identical to its output

ADAS Feature Validation Harness
Python, C++, pybind11, GoogleTest, pytest, highway-env

- Validated three driver-assistance features with a harness built on a **C++** core with CMake and pybind11, backed by **143** pytest and GoogleTest tests across **29** scenarios, and exposed a sensor fusion aliasing bug by checking results against derived values, correcting time-to-collision from **0.746s** to **1.493s**, the exact **2x** the math predicted

LIDNeRF: Language-Guided NeRF Editing
PyTorch, CLIP, Lang-SAM, InstructDiffusion, NeRFStudio, CUDA, Kubernetes

- Developed an automated **NeRF**-editing pipeline combining **CLIP**-guided Lang-SAM segmentation with diffusion models driven by natural language prompts, lifting spatial mapping accuracy from **67% to 82%** for a co-authored **IJIIT 2025** paper, and cut training from **9 to 3 hours** with custom **CUDA** kernels in C++ and GPU memory tuning for training jobs on Kubernetes

TECHNICAL SKILLS

Languages & Frameworks: Python, C++, TypeScript, JavaScript, SQL, Java, React, Node.js, Django REST Framework, FastAPI
AI & ML: PyTorch, CUDA, ONNX, Temporal Fusion Transformer, CLIP, NeRF, Claude Code, OpenAI Codex, MCP
Backend & Data: Microservices, REST/GraphQL APIs, OpenAPI, Postman, SimpleJWT, MySQL, PostgreSQL, Redis
Cloud & Mobile: AWS (EC2, S3, Lambda, CloudWatch), Docker, Kubernetes, Linux/Bash, Git, CI/CD, iOS, Android, Capacitor
Testing & CS Core: GoogleTest, pytest, TDD, CMake, pybind11, Verilator, RISC-V, Data Structures and Algorithms, System Design